

Supplementary tables

Device time and error statistics of every configuration timed or measured on each collection, from the records in **results/fastsar** of the study repository. The column *Rings* gives the error in four concentric regions, the quarters of the half-width from the scene center outward. Configurations are labeled as in the paper.

Table 1: Every configuration of the release records, Panama Canal: time in seconds (host memory to host memory), with the seconds per image of the loop in parentheses where a loop was run; error and largest pixel difference (dB), 5 by 5 coherence statistics, amplitude and phase statistics over the brighter half of the pixels, and the error in four rings by distance from the scene center (quarters of the half-width, dB). A dash in the time column marks a configuration whose image was measured but which was not timed from host memory to host memory; a dash in the other columns, one timed without saving its image.

Device	Configuration	Time	Error	Max	Coh mean	Coh p0.1	Coh min	Amp p99	Amp max	Ph p99	Ph max	Rings
L4	Exact BP, cubic interpolation, 8× oversampled	24.1 (25.6)	-67.4	-72.1	1.0000	1.000	0.998	0.01	0.6	0.1	4	-67, -67, -68, -67
L4	Exact BP, linear interpolation, 8× oversampled	20.3 (20.0)	-56.9	-62.2	1.0000	1.000	0.923	0.03	2.5	0.2	15	-57, -56, -57, -57
L4	Exact BP, linear interpolation, 16× oversampled	26.9 (27.1)	-80.5	-88.8	1.0000	1.000	1.000	0.00	0.1	0.0	0	-80, -81, -81, -80
L4	Factorized BP, float16 throughout, CUDA kernels	3.5 (3.5)	-58.9	-64.7	1.0000	1.000	0.932	0.03	4.2	0.2	20	-58, -59, -59, -59
L4	Factorized BP, float32, CUDA kernels	3.8 (3.8)	-59.9	-65.0	1.0000	1.000	0.931	0.02	4.2	0.1	20	-60, -60, -60, -60
L4	Polar format, float32, gather	3.0 (3.1)	-32.1	-40.3	0.9994	0.986	0.680	0.30	20.7	6.1	70	-40, -37, -35, -30
TPU v5e	Factorized BP, single-pass products, fused kernels	3.6 (2.5)	-46.2	-65.0	0.9999	0.998	0.674	0.16	10.9	1.1	106	-45, -46, -46, -46
TPU v5e	Factorized BP, three-pass products, fused kernels	5.2 (4.0)	-57.9	-64.8	1.0000	1.000	0.931	0.03	4.2	0.2	20	-58, -58, -58, -58
TPU v5e	Polar format, float32, gather	103.4 (103.3)	-32.1	-40.3	0.9994	0.986	0.680	0.69	20.7	6.1	70	-40, -37, -35, -30
TPU v6e	Factorized BP, single-pass products, fused kernels	2.6 (1.6)	-46.2	-64.8	0.9999	0.998	0.674	0.16	10.4	1.1	106	-45, -46, -46, -46
TPU v6e	Factorized BP, three-pass products, fused kernels	3.1 (2.1)	-57.8	-64.5	1.0000	1.000	0.931	0.03	4.2	0.2	20	-58, -58, -58, -58
TPU v6e	Polar format, float32, gather	110.2 (110.2)	-32.1	-40.3	0.9994	0.986	0.680	0.69	20.7	6.1	70	-40, -37, -35, -30
CPU	Exact BP, cubic interpolation, 8× oversampled	433.5	-67.4	-72.1	1.0000	1.000	0.998	0.01	0.6	0.1	4	-67, -67, -68, -67
CPU	Exact BP, linear interpolation, 8× oversampled	292.1	-56.9	-62.2	1.0000	1.000	0.923	0.03	2.5	0.2	15	-57, -56, -57, -57
CPU	Factorized BP, float32, JAX program	224.5	-58.3	-64.8	1.0000	1.000	0.931	0.03	4.2	0.2	20	-58, -58, -58, -58
CPU	Factorized BP, float32, C++ kernels	11.3 (11.3)	-60.0	-65.0	1.0000	1.000	0.931	0.02	4.2	0.1	20	-60, -60, -60, -60
CPU	Polar format, float32, gather	11.4 (11.4)	-32.1	-40.2	0.9994	0.986	0.679	0.69	20.1	6.1	70	-40, -37, -35, -30

Table 2: Every configuration of the release records, Melbourne: time in seconds (host memory to host memory), with the seconds per image of the loop in parentheses where a loop was run; error and largest pixel difference (dB), 5 by 5 coherence statistics, amplitude and phase statistics over the brighter half of the pixels, and the error in four rings by distance from the scene center (quarters of the half-width, dB). A dash in the time column marks a configuration whose image was measured but which was not timed from host memory to host memory; a dash in the other columns, one timed without saving its image.

Device	Configuration	Time	Error	Max	Coh	Coh mean	Coh p0.1	Coh min	Amp p99	Amp max	Ph p99	Ph max	Rings
L4	Exact BP, cubic interpolation, 8x oversampled	26.6 (26.7)	-67.2	-71.3	1.000	1.0000	1.000	0.998	0.01	1.1	0.1	7	-67, -67, -67, -68
L4	Exact BP, linear interpolation, 8x oversampled	20.6 (20.3)	-56.8	-61.3	1.000	1.0000	1.000	0.958	0.04	8.2	0.3	22	-57, -57, -57, -57
L4	Exact BP, linear interpolation, 16x oversampled	27.8 (28.1)	-81.7	-86.7	1.000	1.0000	1.000	1.000	0.00	0.1	0.0	0	-83, -81, -81, -81
L4	Factorized BP, float16 throughout, CUDA kernels	3.6 (3.6)	-58.5	-59.0	1.000	1.0000	1.000	0.930	0.04	2.6	0.2	23	-59, -58, -58, -58
L4	Factorized BP, float32, CUDA kernels	3.9 (4.0)	-59.4	-58.6	1.000	1.0000	1.000	0.930	0.03	2.6	0.2	23	-60, -59, -59, -59
L4	Polar format, float32, gather	3.1 (3.1)	-33.5	-46.7	0.9997	0.9997	0.994	0.873	0.52	23.9	4.1	180	-42, -39, -36, -30
TPU v5e	Factorized BP, single-pass products, fused kernels	3.7 (2.5)	-46.2	-61.8	1.0000	1.0000	0.999	0.859	0.18	12.2	1.2	172	-47, -46, -46, -46
TPU v5e	Factorized BP, three-pass products, fused kernels	5.5 (4.2)	-57.9	-57.6	1.0000	1.0000	1.000	0.930	0.04	2.6	0.3	23	-58, -58, -58, -58
TPU v5e	Polar format, float32, gather	68.2 (68.3)	-33.5	-46.7	0.9997	0.9997	0.994	0.873	0.52	23.9	4.1	180	-42, -39, -36, -30
TPU v6e	Factorized BP, single-pass products, fused kernels	2.6 (1.5)	-46.2	-59.5	1.0000	1.0000	0.999	0.859	0.18	11.7	1.2	173	-47, -46, -46, -46
TPU v6e	Factorized BP, three-pass products, fused kernels	3.1 (2.0)	-57.7	-56.2	1.0000	1.0000	1.000	0.930	0.04	2.6	0.3	23	-57, -57, -57, -58
TPU v6e	Polar format, float32, gather	106.5 (106.5)	-33.5	-46.7	0.9997	0.9997	0.994	0.873	0.52	23.9	4.1	180	-42, -39, -36, -30
CPU	Factorized BP, float32, C++ kernels	12.8 (12.6)	-59.5	-58.7	1.0000	1.0000	1.000	0.930	0.03	2.6	0.2	23	-60, -59, -60, -59
CPU	Polar format, float32, gather	11.9 (12.1)	-33.5	-46.7	0.9997	0.9997	0.994	0.873	0.52	23.8	4.1	179	-42, -39, -36, -30

Table 3: Every configuration of the release records, Iowa farmland: time in seconds (host memory to host memory), with the seconds per image of the loop in parentheses where a loop was run; error and largest pixel difference (dB), 5 by 5 coherence statistics, amplitude and phase statistics over the brighter half of the pixels, and the error in four rings by distance from the scene center (quarters of the half-width, dB). A dash in the time column marks a configuration whose image was measured but which was not timed from host memory to host memory; a dash in the other columns, one timed without saving its image.

Device	Configuration	Time	Error	Max	Coh mean	Coh p0.1	Coh min	Amp p99	Amp max	Ph p99	Ph max	Rings
L4	Exact BP, cubic interpolation, 8x oversampled	18.1 (18.0)	-70.7	-73.1	1.0000	1.000	1.000	0.01	0.3	0.0	1	-71, -70, -71, -71
L4	Exact BP, linear interpolation, 8x oversampled	12.6 (12.8)	-60.5	-63.3	1.0000	1.000	0.994	0.02	1.2	0.1	8	-60, -60, -61, -60
L4	Exact BP, linear interpolation, 16x oversampled	17.1 (17.3)	-82.8	-86.2	1.0000	1.000	1.000	0.00	0.0	0.0	0	-84, -83, -83, -83
L4	Factorized BP, float16 throughout, CUDA kernels	2.3 (2.3)	-59.1	-63.6	1.0000	1.000	0.978	0.02	0.9	0.1	7	-59, -59, -59, -59
L4	Factorized BP, float32, CUDA kernels	2.5 (2.5)	-60.2	-63.7	1.0000	1.000	0.978	0.02	0.9	0.1	7	-60, -60, -60, -60
L4	Polar format, float32, gather	2.3 (2.3)	-35.4	-35.4	0.9998	0.998	0.973	0.33	9.6	2.7	38	-40, -39, -37, -33
TPU v5e	Factorized BP, single-pass products, fused kernels	2.5 (1.7)	-46.3	-54.7	1.0000	1.000	0.984	0.10	3.0	0.6	13	-47, -46, -46, -46
TPU v5e	Factorized BP, three-pass products, fused kernels	3.5 (2.7)	-58.8	-63.6	1.0000	1.000	0.978	0.02	1.1	0.1	7	-59, -59, -59, -59
TPU v5e	Polar format, float32, gather	48.3 (48.3)	-35.4	-35.4	0.9998	0.998	0.973	0.33	9.6	2.7	38	-40, -39, -37, -33
TPU v6e	Factorized BP, single-pass products, fused kernels	1.8 (1.1)	-46.3	-54.9	1.0000	1.000	0.984	0.10	3.0	0.6	13	-47, -46, -47, -46
TPU v6e	Factorized BP, three-pass products, fused kernels	2.2 (1.4)	-58.9	-63.7	1.0000	1.000	0.978	0.02	1.1	0.1	7	-59, -59, -59, -59
TPU v6e	Polar format, float32, gather	61.1 (61.1)	-35.4	-35.4	0.9998	0.998	0.973	0.33	9.6	2.7	38	-40, -39, -37, -33
CPU	Factorized BP, float32, C++ kernels	7.3 (7.5)	-60.2	-63.7	1.0000	1.000	0.978	0.02	0.9	0.1	7	-61, -60, -60, -60
CPU	Polar format, float32, gather	8.4 (8.5)	-35.4	-35.4	0.9998	0.998	0.973	0.33	9.5	2.7	38	-40, -39, -37, -33